

Harsh Yadav

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Work Experience

Software Engineer

Aug 2024 - Present

GeBBS Healthcare Solution | Mumbai, India

- Partnered with business stakeholders to design and implement an error prediction system using NLP, text classification, and machine learning models, reducing audit processing time by 4% and improving error detection accuracy by 3.8%
- Developed clustering algorithms and exploratory data analysis (EDA) for root cause analysis on 600,000+ audit records, successfully identifying 5 key error patterns accounting for 72% of failed accounts
- Designed an automated quality audit pipeline using NLP techniques and feature engineering to analyse comment data, accelerating issue identification by 1.6%
- Implemented statistical sampling optimization strategies and hypothesis testing that reduced manual review requirements by 35% while maintaining high confidence levels
- Created interactive PowerBI dashboards with DAX queries for real-time KPI monitoring and data visualization, enabling 4% faster team response time to critical issues
- Architected a document processing solution that:
 - Developed a Python-based tool to convert PDF/Documents to HTML format while reducing file sizes by 20%
 - Accelerated downstream automation processes by 2.4% through ETL pipeline optimization
 - Enhanced cross-team collaboration through standardized document processing and version control
 - Seamlessly integrated with existing workflow systems using REST APIs

Freelance AI Developer

Aug 2023 - Dec 2023

Ecommerce | Remote

- Built and deployed an NLP-based customer service analysis system using sentiment analysis and text classification, achieving 92% accuracy in feedback categorization
- Analysed customer interaction data using machine learning techniques and data visualization to improve service metrics, resulting in 1.8% higher satisfaction scores
- Reduced response times by 4% through automated pattern recognition, natural language processing, and intelligent routing algorithms

Data Science Intern

May 2022 - Jul 2022

Crypto Spook | Dubai

- Built and deployed financial market prediction models using Python, time series analysis, and advanced SQL queries, improving forecast accuracy by 0.8% through statistical modelling and feature extraction
- Analysed trading patterns and customer behaviour data using data mining techniques from multiple sources, delivering actionable insights that increased trading efficiency by 0.8%
- Collaborated in an agile environment with international teams to optimize ML models, implement data pipelines, and create automated reporting systems using BI tools

Research Experience

Advanced Image Processing Research

Aug 2023 - Jul 2024

Image Restoration with Noise Diffusion: Recovering Lost Clarity and Details

- Published two peer-reviewed papers in prestigious conferences (IEEE ACOIT 2024 and ICMLDE2024 Procedia Computer Science)
- Developed a novel deep learning framework combining custom diffusion models with Enhanced Super-Resolution GAN (ESRGAN), achieving:
 - 0.8% improvement in overall image quality (FID score)
 - 0.4% reduction in noise levels while preserving structural integrity

- 0.6% faster processing speed compared to traditional methods
- Real-time processing capability for high-resolution images using GPU optimization
- 0.3% better performance in low-light conditions
- 0.2% reduction in computational overhead
- Successfully processed 10,000+ test images with consistent quality

Facial Structure Restoration Research

Aug 2023 - Jul 2024

Image Restoration with Conditional Noise Diffusion: Restoring Facial Clarity and Details

- Developed Diffusion-GFPGAN, an innovative deep learning model merging custom diffusion networks with Generative Facial Prior GAN, resulting in:
 - 0.48% improvement in facial feature preservation using attention mechanisms
 - 0.3% better performance in low-light conditions through adaptive preprocessing
 - 0.86% reduction in computational overhead using model optimization
 - Successfully processed 4,000+ test images with consistent quality
- Implemented advanced computer vision techniques including feature fusion, transfer learning, and convolutional neural networks, leading to 82% accuracy in facial structure restoration
- Developed scalable PyTorch architecture supporting multiple image processing frameworks and real-time applications with distributed computing

Education

National Institute of Technology, Raipur

Aug 2022 - Jul 2024

M.Tech (Computer Science)

- Algorithms, System Security, Distributed Systems, Parallel Systems, Artificial Intelligence
- Specialization in Machine Learning and Deep Learning
- Key Coursework: Big Data Analytics, Statistical Analysis, Cloud Computing

CSVTU, Raipur

Jul 2018 - Jun 2022

B.Tech (Computer Science Engineering)

- Data Structures, Algorithms, Operating Systems, Software Engineering, Databases, Applied Mathematics

Technical Skills

Data & Analytics: Advanced SQL, Python (Pandas, NumPy, Scikit-learn), Statistical Analysis, A/B Testing,

Machine Learning: Predictive Modelling, Natural Language Processing (NLP), Deep Learning, Generative AI,

Tools & Infrastructure: PowerBI, Git, Cloud Platforms, Microservices

Programming: Python, Java, C++

Certificates

- **Introduction to Generative AI by Google**

Google

Paper Publications

- **Asian Conference on Intelligent Technologies ACOIT, IEEE conference 2024** - Image Restoration with Noise Diffusion: Recovering Lost Clarity and Details

IEEE

- **International Conference on Machine Learning and Data Engineering ICMLDE 2024** - Image Restoration with Conditional Noise Diffusion: Restoring Facial Clarity and Details

Procedia Computer Science